Matrix discrepancy and optimization

Global synchronization of a random cubic graph

Difficulty: challenging **Importance:** interesting to the community

Status: Solved

Provenance: source-stated conjecture, with the phase domain written explicitly as a torus.

Independently reviewed resolution - 2026-09-11

Negative resolution, Theorem 1. For a uniformly random labelled simple cubic graph on an even number of vertices, the probability that every local minimum of the homogeneous Kuramoto energy is synchronized tends to **zero**, rather than one. With high probability a nonsynchronized local minimum has edge cosines at least $1/32$ and Hessian at least $(1/320)I$ on the mean-zero subspace. The full analytic argument and the primary random-graph inputs passed two independent agent reviews. The graph model, torus topology and quantification over every local minimum are unchanged.

Author: Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Complete manuscript, deterministic review, probabilistic review. See the submission record for provenance and missing-file limitations. This is independent agent verification, not external human peer review or formal proof-assistant certification. No novelty or priority claim is made.

The difficulty, importance and rating rationale below are historical assessments of the original open target.

Rating rationale: Controlling every local minimum on sparse random cubic graphs requires new landscape analysis beyond existing dense and high-degree results; it informs synchronization and nonconvex optimization.

For each even integer $n \geq 4$, let G_n be uniformly distributed over the labelled simple 3-regular graphs with vertex set $\{1, \dots, n\}$. For a graph G on these vertices, define

$$E_G(\theta) = \sum_{\{i,j\} \in E(G)} (1 - \cos(\theta_i - \theta_j)), \qquad \theta \in (\mathbb{R}/2\pi\mathbb{Z})^n.$$

A phase vector is synchronized if $\theta_i = \theta_j$ modulo 2π for every pair i, j. A local minimum is understood with respect to the usual topology on the phase torus; it need not be strict.

Is it true that

$$\lim_{\substack{n\to\infty \\ n \text{ even}}} \mathbb{P}\Big(\text{every local minimum of } E_{G_n} \text{ is synchronized}\Big) = 1?$$

The energy is a sparse graph optimization problem whose derivatives involve weighted graph Laplacians. The conjecture asks whether a typical graph of the smallest plausible fixed regular degree has no nonsynchronized local minima.

References

1. A. S. Bandeira, A. Kireeva, A. Maillard, and A. Rödder, *Randomstrasse101: Open Problems of 2024*, arXiv:2504.20539 (2025), Definition 2.1 and Conjecture 3. Paper.
2. P. Abdalla, A. S. Bandeira, M. Kassabov, V. Souza, S. H. Strogatz, and A. Townsend, *Expander graphs are globally synchronizing*, Advances in Mathematics 488 (2026), article 110773; arXiv:2210.12788v4, introduction and Section 7. The random regular result requires substantially larger fixed degree. Paper.
3. V. Jain, C. Mizgerd, and M. Sawhney, *The random graph process is globally synchronizing*, arXiv:2501.12205 (2025), Theorem 1.2: the connectivity hitting-time result for a different random graph model. Paper.
4. S. Lepsveridze and S. Zhang, *Graphs with connectivity* $3/4-\varepsilon$ *are globally synchronizing*, (2026), dense minimum-degree result. Paper.

Status check — 2026-09-10

checked the current records 2504.20539v1 (2025-04-29), 2210.12788v4 (2026-01-19), 2501.12205v1 (2025-01-21), and 2608.20010v1 (2026-08-20), and searched “random cubic globally synchronizing 2026” and “random regular graphs degree 3 conjecture”. The large-degree, random-graph-process, and dense minimum-degree results do not establish the stated cubic-graph assertion. No proof, counterexample, or relevant withdrawal was found. The source’s angular variables give the torus used here; its isolated sphere notation is not imposed on the phases.

Audit update (2026-09-10): Rechecked the 2025 cubic-graph conjecture and August 2026 dense-degree theorem, and searched for later cubic synchronization results. The dense threshold result concerns a different graph regime and does not resolve this target. This is a bounded literature check, not a proof that no solution exists.